

Modelling, Analysis & Classification of the Pima Indians Diabetes Mellitus Dataset

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Abstract—This paper looks at the effectiveness of different imputation techniques in supporting binary classification performance of the Logistic Regression, J48 decision tree and KNN classifiers in the context of decision support for the healthcare sector where accurate predictive models have the potential to improve patient outcomes by promoting access to care. Processing methodologies are assessed with reference to the relevant Accuracy and False Positive metrics. Analysis and comparison of the results suggests that imputation alone can likely impact classifier performance only marginally in situations where the data set is of sufficient size.

Index Terms—Healthcare, Imputation, Interpretability, Classification, Missing Data.

I. INTRODUCTION

Diabetes mellitus is a high-incidence condition that can be highly treatable with contemporary monitoring and medication solutions. Un-diagnosed or untreated diabetes can be extremely detrimental to the health and quality of life of affected individuals. Long term complications are caused by progressive damage to organs due to improperly regulated blood sugar and includes kidney failure, miscarriages and blindness, and in more extreme situations sufferer's can fall into a coma which will be fatal if untreated.

As such there is significant interest in early diagnosis to put treatment plans in place, which can effectively reduce the impact of the condition, particularly as most clinical diagnoses of diabetes occur in response to tests conducted for other purposes [1]. This suggests there would be utility in a method for early screening on the basis of patient healthcare records as this would reduce the time between incidence and diagnosis. In pursuit of this there have been a number of studies that have attempted to find accurate inference models from patient data. This paper juxtaposes two separate approaches for pre-processing and modelling a diabetes data set to produce trained classifiers that can predict cases of diabetes when presented with previously unseen data. We refer to each separate methodology as 'Method Set A' or 'B' throughout.

The analysed dataset was collected from observation of the Pima Native American population living around Southern Arizona and originally published by the US National Institute of Diabetes and Digestive and Kidney Diseases who conducted a multi-year continuous study to investigate the unusually high diabetes incidence rates in the Pima community. Properly anonymised prior to publication, it has been the subject of numerous analysis studies targeting

machine learning outcomes, and as such represents a well-validated resource for exploration.

The Pima dataset contains 768 instances, comprising 8 predictor features and one binary output class, with the value 1 indicating diabetic and the value 0 indicating non-diabetic. No distinction is made around gradation i.e.(pre-diabetes). All participants were female & over the age of 21 for data to be included.

The primary focus of this exploratory paper is to:

- Assess the predictive power of different classification algorithms applied to this data set.
- Assess the ability of different data-imputation techniques in improving classification performance.
- Attempt to identify clinically relevant inferences regarding the features with the most predictive power in identifying diabetes cases.

II. LITERATURE REVIEW

While reviewing research relating to this data set, it was observed that many studies did not identify or address missing values present in the data due to missing values being erroneously encoded as zero, which prompted exploration of the techniques around handling such cases. Pearson 2006 [2] demonstrated that the treatment of NA values strongly effected statistical properties of the data set, which affects classifier performance.

Chang et al [1] used the Pima Native American data set to assess the performance of three interpretable supervised classifiers (J48 decision tree, random forest classifier and Naive Bayes). The intent was to do so in the context of the 'Internet of Medical Things'(IOMT) which the authors assert is especially sensitive to concerns about interpretability of the resulting models due to the crucial role trust plays in the healthcare sector in providing proper care. The ability of the chosen three models to produce accessible insight into their internal decision making process makes them good candidates for medical applications, as they can mitigate these concerns to some extent. This theme of interpretable models is topical and of broader concern beyond the healthcare sector, with concern being recently expressed over the opacity of algorithmic modelling in domains such as social media [3]. Their proposed architecture for a ML enhanced IOMT e-diagnosis system proposed using the best classifier of the three to produce an indicative preliminary diagnosis for doctors as a form of decision support.

The researchers did several rounds of pre-processing, initially employing median value imputation to deal with invalid zero values for relevant features before applying feature selection methods in later rounds using PCA and k-means clustering). The results obtained were mixed, with models trained on a reduced feature set comprising (glucose, BMI, and age) performing marginally better across the performance metrics.

III. METHODOLOGY

A. Data Analysis

1) *Initial Observations:* For both approaches, analysis of the data set started with preliminary statistical summarisation and visualisation of the feature set (Fig 1). From this a list of initial observations was drawn up to help focus further investigation. These joint observations included:

- A substantial & likely erroneous proportion of zero values in the SkinThickness and Insulin features, with BloodPressure, BMI and Glucose also being affected but to a lesser degree.
- A marked difference in the distribution of glucose when faceted by outcome.
- A large number of statistical outliers were observed for several of the features with Insulin and Skin-Thickness possessing a couple of extreme examples.

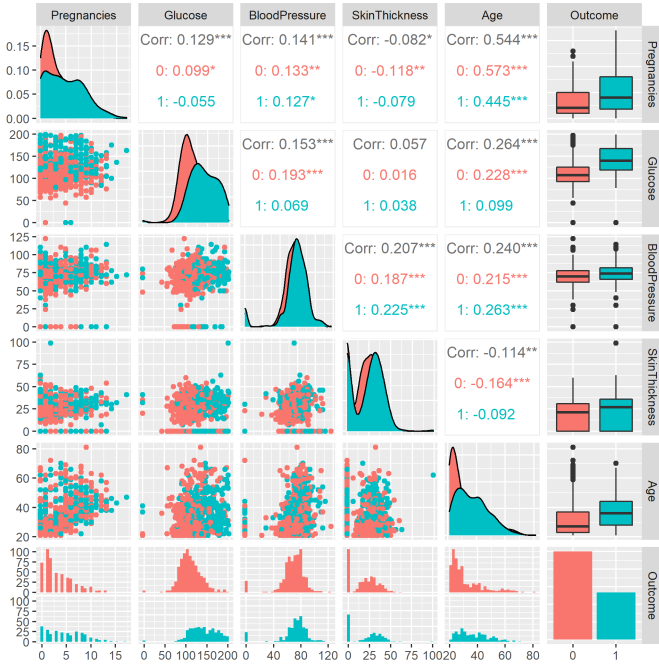


Fig. 1. Initial survey of the data-set (Reduced list of features for display)

Before beginning manipulation of the data set repeatability was ensured by setting a global seed parameter. As observed above, there were a large number of '0' values recorded for several of the features (Pregnancies, Glucose, Blood Pressure, Insulin, SkinThickness, BMI and Pedigree Function). Application of domain knowledge determined that for these (excluding pregnancies) a zero value was outside

of physiological range and could therefore be regarded as erroneous or missing data. To help quantify this and gauge how much potential impact this might have on the data set the zero values were transformed to 'NA' values and visualised using a variety of plots from the 'MICE' and 'Naniar' packages. From this it became clear that a problematic proportion of the Insulin and SkinThickness features were absent (49% & 30% respectively), with a much smaller proportion of values missing for the BloodPressure, BMI and Glucose predictors. Additionally there was discussion about methods to deal with potential 'inliers' present in the Pregnancies feature, whereby missing values had been improperly entered as 0 and were therefore indistinguishable from correctly obtained data.

There are several accepted options when dealing with large volumes of missing data, but what is appropriate depends on the nature of the missing values, with three types posited by Little & Rubin [4], namely Missing Completely at Random (MCAR), Missing at Random (MAR), Not Missing at Random (NMAR). Observation of a trend between the missing values - that they become more frequent as age of participants increases can be taken as indication that the missing data is not MCAR and should be scrutinised further (Fig 2), this was corroborated by using Little's MCAR test statistic ($p = <0.05$).

There is no definitive method for determining if missing values are MAR or NMAR aside from collecting the data, so we proceed under the assumption that they are MAR, which permits the deletion of features that have a large proportion of missing values [5], but otherwise advises imputation of substitute values. Similarly for the potential inliers mentioned above there is no ready mechanism for identification and analysis was continued under the assumption that no values were missing.

Initial boxplots of each feature showed a large number of outliers above the standard threshold of the 75 percentile + $1.5 \times \text{IQR}$. This can be a sign of skewed distributions and indeed this turned out to be the case for several of the features, with Age being the clearest example. Further analysis diverged between the method sets.

Method Set A: The correlation matrix is generated as shown in Fig.2 which shows variables having a strong correlation with one another. This is important to know since several machine learning methods, such as linear and logistic regression, might perform poorly when our data contains strongly correlated input variables. Skin thickness and insulin showing negatively correlated to the age and pregnancies but there are no negative correlated variables to outcome. Additionally, no single predictor revealed a strong positive linear relationship with a correlation of at least +0.70.

Method Set B: Having identified that skewed distributions were generating large numbers of candidate 'outliers', these values were manually reviewed for plausibility as determined by domain knowledge. As a result of this it was decided that

Correlation Between Variables and outcome

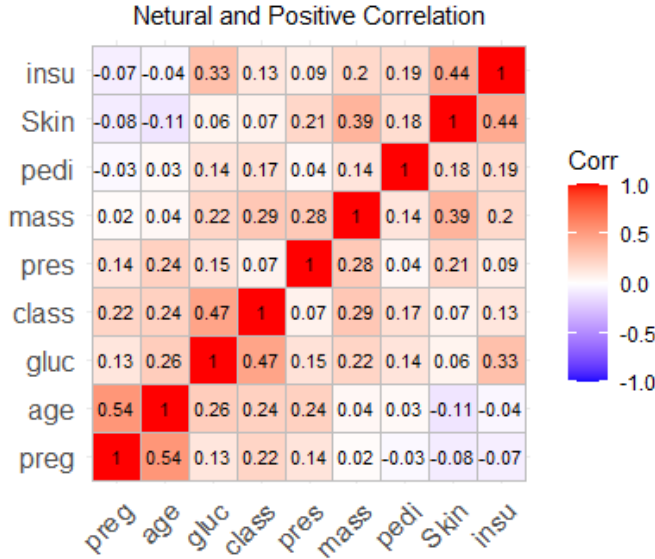


Fig. 2. Correlation Between Variables and Outcome

the threshold value for marking outlier values for removal should be based on a $3 \times \text{IQR}$, as this is a standard approach to identifying extreme outliers.

Quartile-Quartile (QQ) plots were used support this observation. QQ plots are a useful tool for observing distributional anomalies. Non-normal distributions have a noticeable deviation from the 45 degree reference line—here the difference is especially evident between the Age and SkinThickness variables, with Age evidently not following a normal distribution, with the horizontal artefact around age 20 due to the artificial age cut off imposed by only including those over the age of 21 in the study (figure 3).

It was observed that the incidence of obesity amongst the diabetic population sample was considerably higher than that of the non-diabetic population (fig 7).

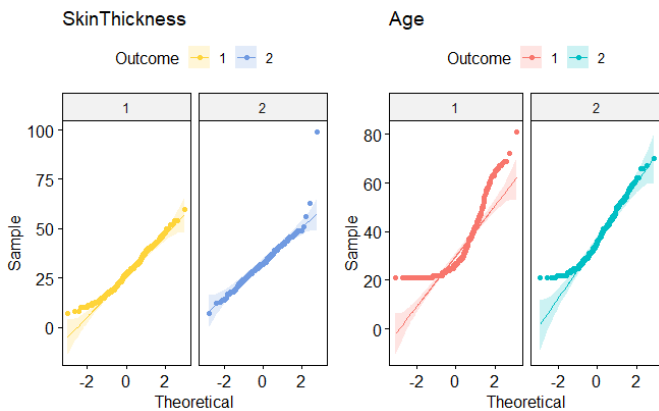


Fig. 3. Q-Q plot Indicating the normality in distribution of the Age and SkinThickness features, faceted by outcome. Note the pronounced divergence from the reference line in the Age plot.

B. Pre-Processing

1) Handling Missing Values:

Method Set A followed a four stage process:

- 1) Remove Outliers: Initial density curves shows that all features are not normally distributed. So the outliers are recognised by generating box-plots with scatter and are deleted accordingly.
- 2) Find missing values: The instances with any of the following missing value cases are removed:
 - Where 3 or 4 values are missing.
 - Where Insulin and Glucose values missing simultaneously as they are highly correlated.
 - Where Insulin and SkinThickness are jointly missing. This represents roughly 25% of the data set.
- 3) Impute missing values: The remaining missing values

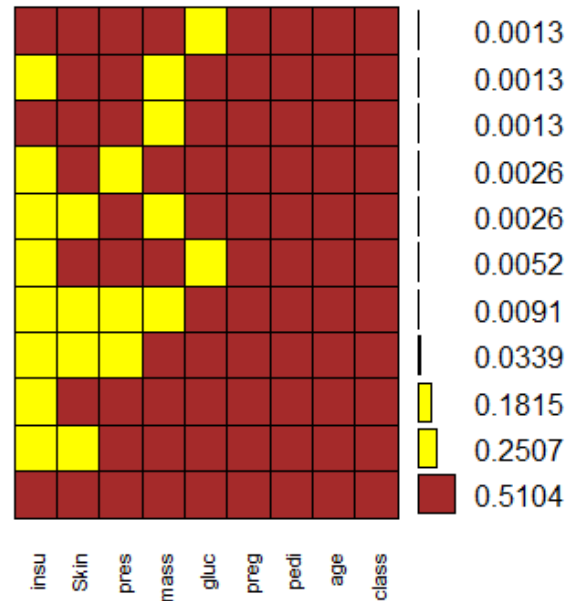


Fig. 4. The pattern of ratio of missing values

are then imputed using a hybrid approach of single and multiple imputation methods - Single Center Imputation from Multiple Chained Equation (SICE) [6] which is an extension of Multivariate Imputation by Chained Equation (MICE) [7]. MICE uses different simulation methods to produce multiple values for the imputation of a single missing value. This approach first uses MICE with Predictive Mean Matching (PMM) and then imputes missing values for numeric attributes by replacing it with the mean of the multiple values produced by MICE. PMM is a general purpose method of imputing missing values by drawing real values sampled from the data [8].

- 4) Remove outliers after imputation: Even though SICE cannot impute "impossible" imputed values as the imputed values are restricted to observed values, the box plot is used to catch outliers missed in previous step.

Following these procedures, there are a total of 522 cases left. The correlation matrix plot is again generated and observed

for the highly correlated features. The top four features correlated with outcome were glucose, insulin, BMI and age.

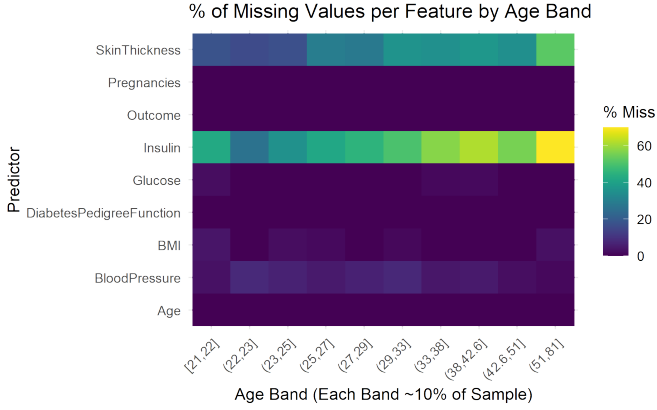


Fig. 5. The proportion of missing values increases with age

Method Set B: Two-way t-tests were performed on all features (ignoring NA values) as an assessment of feature importance. All features were reported to be statistically significantly different at the 95% confidence interval. As such no features were removed on the basis of this.

Two methods were chosen for processing the data so as to allow comparison of the effectiveness of outlier removal. Firstly a baseline 'Minimal Processing' approach was established by simply using median imputation to replace missing values (This was subsequently added to method A to facilitate comparison). Median was chosen to reduce the effect of outliers as no data was removed under this approach. The pre-processing methods for the second approach can be summarised thus:

- Removing extreme outliers, as defined above.
- The removal of the Insulin feature as 50% of the values for this attribute would have to be imputed.
- Subsequent to the above remove cases where 2 attribute values are missing.

After these adjustments the data set was reduced from 9 to 8 features and from 768 instances to 746 (22 Removed). With the Insulin feature and cases of 2 missing attribute values removed, remaining missing values were imputed using the random-forest based missForest algorithm which has been shown to significantly improve classifier performance (AUC) and impute values that avoid compressing statistical measures such as standard error in comparison to simple mean/median imputation [9]. The Nanier packages 'bindShadow' function was used to generate a matrix indicating where values had been imputed for convenient comparison, and plots were generated to view the distribution of imputed values to monitor resultant distributions vs extant data.

Where relevant, predictors were normalised before training models to avoid unit variance affecting "distance based" classifiers such as KNN. After normalisation, the data set was split 70:30 into training and testing partitions.

C. Classification

1) Method Set A: The generated data set was divided into training and testing in a 70:30 ratio. As the dependent variable (Outcome) is a categorical i.e. whether the person has diabetes or not, this classification can be carried out by logistic regression classifier. A binary logistic regression model is used to fit the training data set. The variables influencing the outcome are then identified by p-values and then others are eliminated till all variables are equally influencing. Finally, a logistic regression model of all the following influencing variables is created: plasma glucose concentration, BMI, and pedigree function.

2) Method Set B: The resultant 'minimal pre-processing' and 'processed' versions of the data set had J48 Decision trees were fitted to the training splits and the resultant models were evaluated using the k-fold cross-validation function native to the 'rWeka' function library ($k = 10$). From this the results were compared and graphics generated from the confusion matrices and resultant decision trees. Additional exploration was performed by fitting a k-nearest-neighbours model to both data sets and graphing the accuracy metric resulting from different k-values. This was done to establish consensus among different types of model as to the trend of effect of the pre-processing approach.

IV. RESULTS

1) Analysis: Method Set A: There are medically impossible values given in the dataset for BMI and Skin Thickness as 0. Ignoring these values, it was observed that majority of women with low values of Skin Thickness and BMI did not have Diabetes.(Fig. 6)

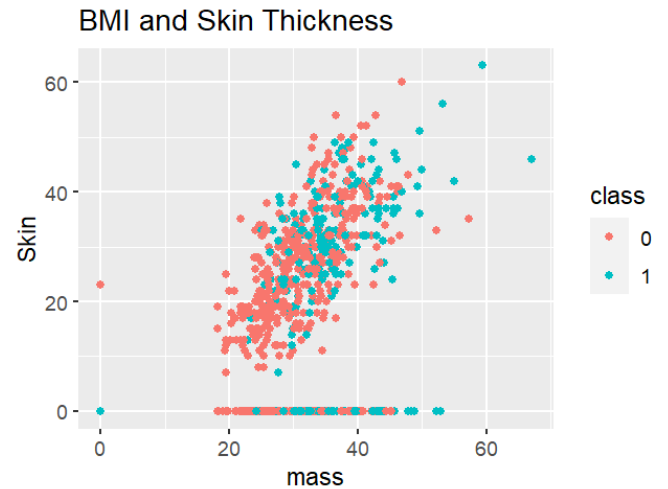


Fig. 6. Non-diabetic women with low values of BMI and Skin Thickness

Method Set B:

The pairs plot represented by figure 5 was used for bi-variate relationship screening. From this it was noted that there were no non-linear relationships between variables, which simplifies analysis. The distribution represented in the glucose feature was noticeably skewed right for the diabetic class,

and the corresponding box-plot indicates that it is likely that the confidence intervals for the means of each group do not overlap.

One observation that might have clinical utility is obtained by stratification of the study participants by BMI, into two groups, those with a BMI under 30 and those that are 30 or above. Plotting the resultant proportions reveals a marked increase in diabetic participants in the over 30 group (fig 7)

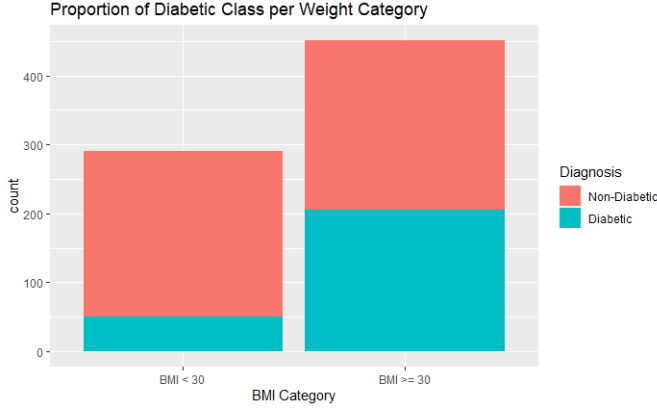


Fig. 7. People with BMI 30 show a greater incidence of Diabetes

2) *Pre-Processing*: The impact of our pre-processing approaches are summarised in the appendix (table III). Key observations from this include:

- Despite being processed with quite distinct methods, the statistical profiles of the resulting data-sets was quite similar for Method Set A and B.
- Both pre-processing approaches differed from the raw data set, with feature statistics such as kurtosis, range and standard deviation all demonstrating the effect of outlier removal. For example the kurtosis metric for blood pressure decreased from >5 to <1.
- The effect of imputing missing values was again particularly evident in the features with a high proportion of missing values. For example for Insulin and SkinThickness features the mean increased from 79 to 146 and 20.54 to 28.89 respectively, for method set A.
- For features that were not badly affected by missing values the effect on calculated statistics was, as expected fairly minimal. When tabulated the degree to which this was the dominant character of the data-set became more apparent, and potentially explains why imputation methods did not have a large effect on the performance of the classifier.

3) *Classification*: Various metrics were used to compare classification performance. Accuracy, as defined by dividing the total number of correct predictions by the total number of predictions, is a typical indicator of model quality and the more robust estimation statistic of the 95% CI for it is used here. Additionally in the healthcare context, the type two error or false negative rate (FNR) is the most impactful metric as it is the one with the worst health impact. As such FNR was also used to assess model performance.

TABLE I
METHOD SET A: LOGISTIC REGRESSION MODEL RESULTS

Classifier (feature selection)	Accuracy	95% CI	Sensitivity	FNR%
LR (Minimal Processing)	0.7917	(0.73, 0.85)	0.9141	23
LR (All)	0.8141	(0.74, 0.87)	0.9057	24
LR (plas+pedi+mass+Age)	0.8141	(0.74, 0.87)	0.8962	26
LR (plas+pedi+mass)	0.8269	(0.75, 0.88)	0.9057	23

TABLE II
METHOD B: MODEL PERFORMANCE MEASURES

Classifier	Accuracy	95% CI	Sensitivity	FNR (%)
J48 (Minimal)	0.7435	(0.68, 0.80)	0.7826	44
KNN(Minimal)	0.7565	(0.70, 0.81)	0.7831	45
J48 (PreProc)	0.7309	(0.67, 0.79)	0.7771	46
KNN (PreProc)	0.7444	(0.68, 0.80)	0.7711	50

Method Set A: In contrast to the correlation matrix analysis, after conducting a logistic regression model, it was discovered that plasma glucose concentration, Body Mass Index, pedigree function were the leading top 3 causes of diabetes in this analysis. The selection of these 3 features resulted into a high accuracy and low FNR.

Method Set B: Performance metrics generated from models trained on both the pre-processed and un-processed data sets were very similar and characterised by high false negative rates throughout (table II). Decision trees calculate attribute importance metrics which were captured for comparison, and visualised by plotting the resultant tree. The three most important predictors were given relative scores like so: Glucose 0.13; Age 0.05, BMI 0.05.

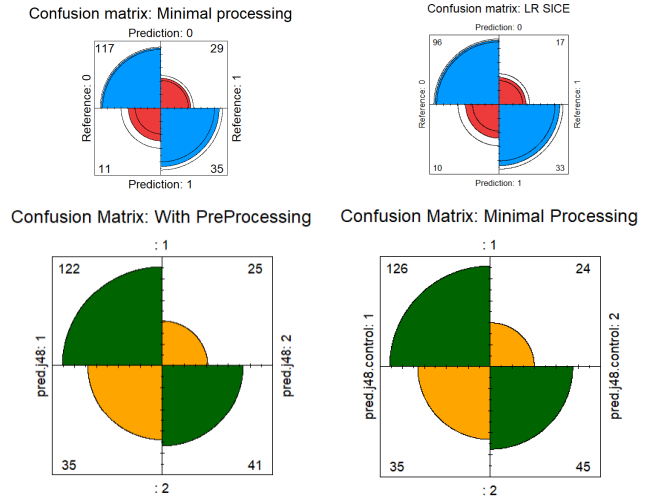


Fig. 8. Top - Method Set A: Logistic regression; Below - Method Set B: J48 - Processed vs Minimal Processing

A p value of 0.90 for insulin means that we cannot reject the null hypothesis that there is no relationship between outcome and Insulin.

V. DISCUSSION

This paper presents two core approaches to the analysis and processing of the Pima Indian data set with the intention

of improving classifier performance. Exploration of the data set was performed separately, however the resulting pre-processing steps produced two data set's that the authors were surprised to observe were very similar to each other, with the two most salient processing steps comprising of the imputation missing values and removing outliers.

A common reference baseline used for binary classification problems is the 'ZeroR' classification accuracy, which essentially represents the proportion of the majority class. In the case of this data set, the ZeroR rate is 66%. The 95% CI lower bound observed for all models exceeds this, so we can say that there is predictive power in the data set. Our stated objective was to ascertain the extent to which pre-processing approaches could improve classifier performance. In method set A, we observe a minor increase across the performance metrics captured in models trained on the pre-processed data set vs the median imputed data set. Further feature selection as outlined seemed to trend toward higher performance however the observed effect was marginal and further testing is needed to confirm.

Similarly for method set B no improvement in performance was observed between the pre-processing approaches across both classifiers. These observations reflect previously obtained results showing that imputation methods alone have minimal impact on classifier performance [9]. The performance demonstrated by the logistic regression classifier was noticeably superior to the decision tree with both improved accuracy and substantially better False Negative Rates. This may be due to a linear separability property of the data set that makes logistic regression intrinsically well suited for application to this data, and indeed recent work on the Pima Indian data set has shown that logistic regression compares favourably to other supervised learning approaches on this data set [10].

The contrast in treatment of instances with missing values may also account for the increased performance observed between approaches, and we think this is more likely. If this is the case, then we interpret it by saying that in removing the instances considered incomplete in Method Set A, some quantity of noisy data was removed, which increased the generalizability of decision boundaries fitted by models trained on this data.

For example, taking the methodology producing the most accurate models in a recent comparative study [11] as an example, Illago et al [12] were more aggressive in removal of 'unclean' data. They removed both the Insulin and SkinThickness features and deleted all remaining values, however this was simply the first step employed in the pre-processing pipeline. Further to this they combined F-Score feature selection with K-means clustering to identify additional instances that were in some sense 'noisy', resulting in a reduced set of 465 instances. This style of approach is common to several papers reporting exceptional classification performance and represents a logical progression of the pre-processing exploration done here [13].

Additionally the data set is a mildly imbalanced one with a roughly 2:1 class ratio. Recent work has demonstrated significant performance increases across a range of classifiers

by balancing the classes using SMOTE approaches based on synthetic minority oversampling techniques (SMOTE) [14]. This may have added relevance as the Pima population was chosen specifically due to it's high incidence rate of diabetes, and so generalising the model building approach to other populations is likely to suffer from more severe imbalances which might need to be addressed.

In terms of potential real world impact, the results here are too weak to consider clinical utility, however the potential for decision support on the basis of systems like this is immediately obvious, particularly in the case of non-invasive screening - for example classifier input could easily assist targeting of informational campaigns like, for example and informational newsletter on the symptoms of diabetes.

VI. CONCLUSION & RECOMMENDATIONS

Overall the pre-processing approaches utilised were demonstrated to have minimal effectiveness in promoting classifier performance. A weakness of this study was to limit exploration of pre-processing approaches largely to imputation techniques. More impactful analysis might have been obtained if the employed approaches had gone further by imposing more rigorous feature selection or transformation techniques as described in the papers mentioned above. An obvious point of further investigation would be in assessing the value of imputation at all, it may simply be situationally appropriate when the size of the data is constrained to the point where removing data will reduce the statistical power of the data set but further work would have to be undertaken to demonstrate this.

VII. CONTRIBUTION

Contributions to this paper were split roughly as denoted by the 'Method Set' annotation throughout, with each author solely contributing to a single method set (psxsj9: Method Set A, psxdf1: Method Set B). Contributions to the writing of the paper, discussion and literature review were approximately equal.

TABLE III
DATA SET STATISTICS POST PRE-PROCESSING

MethodSet	Statistic	Age	BloodPressure	BMI	Ped.Func	Glucose	Insulin	Pregnancies	SkinThickness
-	Kurtosis	0.62	5.12	3.24	5.53	0.62	7.13	0.14	-0.53
A	Kurtosis	0.42	0.75	0.13	0.69	-0.28	3.91	0.7	-0.48
B	Kurtosis	0.64	0.94	0.06	1.76	-0.23	NA	0.13	-0.16
-	Max	81	122	67.1	2.42	199	846	17	99
A	Max	65	110	57.3	1.48	199	600	17	60
B	Max	81	122	57.3	1.73	199	NA	17	60
-	Mean	33.24	69.11	31.99	0.47	120.89	79.8	3.85	20.54
A	Mean	31.4	71.5	32.73	0.48	120.06	146.77	3.56	28.89
B	Mean	33.26	72.4	32.26	0.46	121.03	NA	3.88	28.63
-	Min	21	0	0	0.08	0	0	0	0
A	Min	21	24	18.2	0.09	56	15	0	7
B	Min	21	24	18.2	0.08	44	NA	0	7
-	Q1	24	62	27.3	0.24	99	0	1	0
A	Q1	23	64	27.8	0.26	98	81	1	21
B	Q1	24	64	27.4	0.24	99	NA	1	22
-	Q3	41	80	36.6	0.63	140.5	127.5	6	32
A	Q3	38	80	36.8	0.65	140	182	6	36
B	Q3	41	80	36.4	0.62	140	NA	6	35
-	Std.Dev	11.76	19.36	7.88	0.33	31.97	115.24	3.37	15.95
A	Std.Dev	10.36	12.14	6.58	0.29	30.38	97.7	3.34	9.96
B	Std.Dev	11.74	12.13	6.7	0.3	29.92	NA	3.38	8.91

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